Elementary Analysis

Chapter 3, Section 19

Uniform Continuity
Exercises 19.1–19.12, with every subpart

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 3. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

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19 Uniform continuity

Uniform continuity on a set S means that, for each $\varepsilon > 0$, one positive δ works for *every* pair $x, y \in S$:

$$|x - y| < \delta \quad \Longrightarrow \quad |f(x) - f(y)| < \varepsilon.$$

The choice may depend on ε and on the fixed set S, but not on the chosen points. Restricting a uniformly continuous function to a smaller set preserves this implication, since it reduces the number of pairs to check.

We use Theorem 19.2 for continuous functions on closed bounded intervals and Theorem 19.6 for continuous interval functions with bounded derivative. A bound $|f'| \leq M$ with $M > 0$ gives $|f(x) - f(y)| \leq M\,|x - y|$ by the Mean Value theorem, so $\delta = \varepsilon/M$ works. Derivatives are used only where the stated interval and differentiability hypotheses are satisfied.

19.1 Exercise 19.1: Seven tests for uniform continuity

What we must show. Decide uniform continuity on each specified domain, and justify each answer. In part (e) the function is $1/x^3$, and in part (f) the sine argument is $1/x^2$.

(a) The continuous expression on $[0, \pi]$.

For

$$f(x) = x^{17} \sin x - e^x \cos(3x),$$

the power, sine, exponential, and cosine functions are continuous. Composition gives continuity of $\cos(3x)$, multiplication gives continuity of both products, and subtraction gives continuity of f on $\mathbb{R}$. In particular it is continuous on the closed bounded interval $[0, \pi]$. Theorem 19.2 makes f uniformly continuous there. No estimate involving its complicated derivative is needed.

(b) The cube on $[0, 1]$.

The polynomial x^3 is continuous on $\mathbb{R}$, hence on $[0, 1]$. Theorem 19.2 therefore proves uniform continuity on $[0, 1]$. One can also see a point-independent estimate directly: if $0 \leq x, y \leq 1$, then

$$\left|x^3 - y^3\right| = |x - y| \left|x^2 + xy + y^2\right| \leq 3\,|x - y|\,.$$

Thus $\delta = \varepsilon/3$ works.

(c) The cube on $(0, 1)$.

The function in (c) is the restriction of the uniformly continuous function in (b). The same δ therefore works for all pairs in $(0, 1)$. A domain need not be closed for its function to be uniformly continuous.

(d) The cube on all of $\mathbb{R}$.

For $n \in \mathbb{N}$, choose

$$x_n = n, \qquad y_n = n + \frac{1}{n^2}.$$

Their distance $|x_n - y_n| = 1/n^2$ tends to zero, but expansion of the cube gives

$$y_n^3 - x_n^3 = 3n^2 \frac{1}{n^2} + 3n \frac{1}{n^4} + \frac{1}{n^6} = 3 + \frac{3}{n^3} + \frac{1}{n^6} \geq 3.$$

Fix $\varepsilon_0 = 1$. For any proposed $\delta > 0$, choose n so large that $1/n^2 < \delta$. The corresponding pair is closer than δ, but its function values differ by at least $3 > \varepsilon_0$. This violates Definition 19.1. The cube is not uniformly continuous on $\mathbb{R}$.

(e) The reciprocal cube on $(0, 1]$.

The sequence $x_n = 1/n$ lies in $(0, 1]$ and converges to zero, so it is a Cauchy sequence. For $f(x) = 1/x^3$ its image is $f(x_n) = n^3$, which is unbounded. Every real Cauchy sequence is bounded, so this image is not a Cauchy sequence. If f were uniformly continuous, Theorem 19.4 would make the image of every Cauchy sequence a Cauchy sequence. This contradiction proves that $1/x^3$ is not uniformly continuous on $(0, 1]$.

(f) The rapidly oscillating sine on $(0, 1]$.

Let $f(x) = \sin(1/x^2)$ and choose

$$x_n = \frac{1}{\sqrt{2\pi n + \pi/2}}, \qquad y_n = \frac{1}{\sqrt{2\pi n + 3\pi/2}}.$$

The denominators exceed 1, so both sequences lie in $(0, 1]$. Both tend to zero, whence $|x_n - y_n| \leq x_n + y_n \to 0$. On the other hand,

$$f(x_n) = \sin(2\pi n + \pi/2) = 1, \qquad f(y_n) = \sin(2\pi n + 3\pi/2) = -1.$$

Their images always differ by 2. Taking $\varepsilon_0 = 1$, every proposed $\delta > 0$ is defeated by a sufficiently late pair with $|x_n - y_n| < \delta$. Thus f is not uniformly continuous on this interval.

(g) The damped oscillation on $(0, 1]$.

For $x > 0$ let $f(x) = x^2 \sin(1/x)$, and extend the formula by $F(0) = 0$, $F(x) = f(x)$ for $0 < x \leq 1$. For positive x it is a product and composition of continuous functions. At zero,

$$|F(x) - F(0)| = \left|x^2 \sin(1/x)\right| \leq x^2.$$

Given $\varepsilon > 0$, the choice $\delta = \sqrt{\varepsilon}$ makes this error less than ε whenever $0 \leq x < \delta$. Thus F is continuous on $[0, 1]$. Theorem 19.2 makes F uniformly continuous there, and restriction proves uniform continuity of f on $(0, 1]$.

Conclusion. Parts (a), (b), (c), and (g) are uniformly continuous on their specified sets. Parts (d), (e), and (f) are not.

19.2 Exercise 19.2: Three direct epsilon–delta proofs

What we must show. Verify Definition 19.1 directly for each function, choosing δ independently of the pair of points.

(a) The affine function $3x + 11$ on $\mathbb{R}$.

Let $\varepsilon > 0$ and set $\delta = \varepsilon/3 > 0$. For arbitrary $x, y \in \mathbb{R}$ with $|x - y| < \delta$, cancellation of the constants gives

$$|(3x + 11) - (3y + 11)| = 3\,|x - y| < 3\delta = \varepsilon.$$

This one δ works throughout the unbounded domain.

(b) The square on $[0,3]$.

For $x, y \in [0,3]$, one has $0 \le x+y \le 6$. Factor the difference of squares to get

$$\left|x^2 - y^2\right| = |x-y|\,|x+y| \le 6\,|x-y|\,.$$

For any $\varepsilon > 0$, take $\delta = \varepsilon/6$. Then $|x-y| < \delta$ makes the displayed error less than $6\delta = \varepsilon$, for every such pair.

(c) The reciprocal on $[1/2, +\infty)$.

If $x, y \ge 1/2$, both denominators are positive and $xy \ge 1/4$. Consequently

$$\left|\frac{1}{x} - \frac{1}{y}\right| = \frac{|y-x|}{xy} \le 4\,|x-y|\,.$$

Choose $\delta = \varepsilon/4$. For every pair in the domain with distance less than δ, the function-value distance is less than $4\delta = \varepsilon$.

Conclusion. The respective choices $\varepsilon/3$, $\varepsilon/6$, and $\varepsilon/4$ establish all three uniform-continuity claims directly from the definition.

19.3 Exercise 19.3: Direct proofs for two rational functions

What we must show. Repeat the direct epsilon–delta verification for $x/(x+1)$ on $[0,2]$ and $5x/(2x-1)$ on $[1,+\infty)$.

(a) Subtract with the denominator kept explicit.

For $x, y \in [0,2]$, both $x+1$ and $y+1$ are at least 1. Common denominators and cancellation give

$$\begin{aligned}\left|\frac{x}{x+1} - \frac{y}{y+1}\right| &= \frac{|x(y+1) - y(x+1)|}{(x+1)(y+1)} \\ &= \frac{|xy + x - xy - y|}{(x+1)(y+1)} = \frac{|x-y|}{(x+1)(y+1)} \le |x-y|\,.\end{aligned}$$

Given $\varepsilon > 0$, set $\delta = \varepsilon$. The final bound then becomes less than ε for every pair with $|x-y| < \delta$.

(b) Do the same calculation on the unbounded interval.

For $x, y \ge 1$, one has $2x - 1 \ge 1$ and $2y - 1 \ge 1$. Thus

$$\begin{aligned}\left|\frac{5x}{2x-1} - \frac{5y}{2y-1}\right| &= \frac{|5x(2y-1) - 5y(2x-1)|}{(2x-1)(2y-1)} \\ &= \frac{5\,|-x+y|}{(2x-1)(2y-1)} \le 5\,|x-y|\,.\end{aligned}$$

Set $\delta = \varepsilon/5$. The error is then less than $5\delta = \varepsilon$ whenever $|x-y| < \delta$.

Conclusion. Both functions are uniformly continuous. The direct choices are $\delta = \varepsilon$ in (a) and $\delta = \varepsilon/5$ in (b).

19.4 Exercise 19.4: A uniformly continuous function on a bounded set is bounded

What we must show. Prove the boundedness assertion in (a), without assuming the domain is an interval or is closed, and apply it to part (b).

(a), preparation: account for the empty domain.

If $S = \varnothing$, every bound is a bound on its function values, so the conclusion is immediate. Suppose henceforth that S is nonempty and bounded, and that f is uniformly continuous on S.

(a), contradiction: select values that are too large.

If f were unbounded, then for every integer $n \geq 1$ some $x_n \in S$ would satisfy $|f(x_n)| > n$. Otherwise a bound on all $|f(x)|$ would exist, which is exactly boundedness of f. Since S is bounded, the real sequence (x_n) is bounded. Bolzano–Weierstrass (Theorem 11.5) supplies a subsequence x_{n_j} converging to some real number x_0. It is not necessary that x_0 belong to S.

Every convergent real sequence is Cauchy, so (x_{n_j}) is a Cauchy sequence of points in S. Theorem 19.4 then makes $(f(x_{n_j}))$ a Cauchy sequence, hence a bounded sequence. However, its construction gives

$$\left|f(x_{n_j})\right| > n_j \geq j.$$

Here $n_j \geq j$ because the selected indices are strictly increasing positive integers. The displayed lower bound makes the image sequence unbounded, a contradiction. Thus f is bounded.

(b) Apply the assertion to the reciprocal square.

The set $(0, 1)$ is bounded. For $n \geq 1$ the points $x_n = 1/(n + 1)$ lie in $(0, 1)$, while

$$\frac{1}{x_n^2} = (n + 1)^2 \longrightarrow +\infty.$$

The function $1/x^2$ is therefore unbounded on this bounded domain. If it were uniformly continuous, part (a) would make it bounded. Hence it is not uniformly continuous on $(0, 1)$.

Conclusion. Uniform continuity implies boundedness on every bounded real domain, including domains with missing limit points. This excludes uniform continuity of $1/x^2$ on $(0, 1)$.

19.5 Exercise 19.5: Six domains for trigonometric and reciprocal functions

What we must show. Determine which of the six specified restrictions are uniformly continuous, with the domain included in each argument.

(a) The tangent on $[0, \pi/4]$.

Cosine has no zero in this closed interval. Thus $\tan x = \sin x / \cos x$ is continuous there, and Theorem 19.2 gives uniform continuity.

(b) The tangent on $[0, \pi/2)$.

This domain is bounded, but tangent is unbounded on it. For example, let $x_n = \pi/2 - 1/n$, which lies in the domain. Then

$$\sin x_n \to 1, \qquad \cos x_n \to 0, \qquad \cos x_n > 0.$$

For any $M > 0$, the first convergence gives $\sin x_n > 1/2$ eventually, and the second gives $0 < \cos x_n < 1/(2M)$ eventually. At indices satisfying both estimates, $\tan x_n > M$. Thus tangent is unbounded and cannot be uniformly continuous on this bounded set by Exercise 19.4(a).

(c) The function $(\sin^2 x)/x$ on $(0, \pi]$.

We need the elementary estimate $|\sin t| \le |t|$. It follows from the Mean Value theorem applied to sine between 0 and t: its derivative is cosine, with absolute value at most 1. Exercise 19.8 supplies the more general version. For $x > 0$ this gives

$$0 \le \frac{\sin^2 x}{x} \le \frac{x^2}{x} = x.$$

Extend the function to $[0, \pi]$ by assigning value 0 at zero. The displayed bound proves continuity at zero: if $0 \le x < \varepsilon$, its value has absolute value less than ε. Away from zero the quotient is continuous. The extension is therefore continuous on $[0, \pi]$, hence uniformly continuous by Theorem 19.2. Its restriction to $(0, \pi]$ is uniformly continuous.

(d) The function $1/(x-3)$ on $(0, 3)$.

For $x_n = 3 - 1/n$, one has $2 \le x_n < 3$, so x_n is in the domain, and

$$\frac{1}{x_n - 3} = -n.$$

The function is unbounded on the bounded set $(0, 3)$. Exercise 19.4(a) therefore excludes uniform continuity.

(e) The same function on $(3, +\infty)$.

This domain is unbounded, so unboundedness of the function alone would not settle uniform continuity. Instead take $x_n = 3 + 1/n$. This converges to 3 and is therefore a Cauchy sequence of points in the domain. Its image is

$$\frac{1}{x_n - 3} = n,$$

which is unbounded and hence is not Cauchy. The Cauchy-sequence preservation theorem (Theorem 19.4) excludes uniform continuity.

(f) The same function on $(4, +\infty)$.

For $x, y > 4$, both $x - 3$ and $y - 3$ exceed 1. Thus

$$\left| \frac{1}{x-3} - \frac{1}{y-3} \right| = \frac{|x-y|}{(x-3)(y-3)} \le |x-y|.$$

The choice $\delta = \varepsilon$ works for every pair in this domain. Hence the function is uniformly continuous here.

Conclusion. Parts (a), (c), and (f) are uniformly continuous; parts (b), (d), and (e) are not.

19.6 Exercise 19.6: Uniform continuity without a bounded derivative

What we must show. For $f(x) = \sqrt{x}$, show the derivative is unbounded on $(0, 1]$ while f is uniformly continuous there, and prove uniform continuity on $[1, +\infty)$.

(a), derivative: exhibit arbitrarily large values.

For $x > 0$, elementary differentiation gives $f'(x) = 1/(2\sqrt{x})$. The points $x_n = 1/n^2$ belong to $(0, 1]$ and satisfy

$$f'(x_n) = \frac{1}{2(1/n)} = \frac{n}{2} \longrightarrow +\infty.$$

Thus the derivative is unbounded on this interval.

(a), the function: use the square-root difference estimate.

For $x \geq y \geq 0$, both $\sqrt{x} - \sqrt{y}$ and $\sqrt{x} + \sqrt{y}$ are nonnegative, and the latter is at least the former. Therefore

$$x - y = (\sqrt{x} - \sqrt{y})(\sqrt{x} + \sqrt{y}) \geq (\sqrt{x} - \sqrt{y})^2.$$

Taking nonnegative square roots yields $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$. If $y > x$, interchange the two points to obtain the same estimate. For $\varepsilon > 0$, choose $\delta = \varepsilon^2$. Whenever $x, y \in (0, 1]$ and $|x - y| < \delta$, it follows that

$$|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|} < \sqrt{\delta} = \varepsilon.$$

This establishes uniform continuity despite the unbounded derivative. Theorem 19.6 gives a sufficient condition, not a necessary condition, so there is no contradiction.

(b) Rationalize on the domain bounded away from zero.

For $x, y \geq 1$, one has $\sqrt{x} + \sqrt{y} \geq 2$. Hence

$$|\sqrt{x} - \sqrt{y}| = \frac{|x - y|}{\sqrt{x} + \sqrt{y}} \leq \frac{1}{2} |x - y| .$$

Taking $\delta = 2\varepsilon$ proves uniform continuity on $[1, +\infty)$ directly.

Conclusion. The square root is uniformly continuous on both specified domains. Its derivative is nevertheless unbounded near zero.

19.7 Exercise 19.7: Joining a compact initial interval to a uniform tail

What we must show. Prove part (a)'s extension from a tail to the whole nonnegative half-line, and apply it to square root in part (b).

(a), preparation: choose overlapping intervals.

Let f be continuous on $[0, +\infty)$ and uniformly continuous on $[k, +\infty)$ as in the hypothesis. We only need the part of that tail contained in its domain. Put $K = \max\{k, 0\}$; restriction gives uniform continuity on $[K, +\infty)$. If $k \leq 0$ and the stated tail hypothesis is meaningful, it already includes $[0, +\infty)$, but the argument below also works.

The restriction to $[0, K + 1]$ is continuous on a closed bounded interval, hence uniformly continuous by Theorem 19.2. These two intervals overlap on $[K, K + 1]$. The overlap will handle pairs that are close but lie on opposite sides of the cutoff.

(a), verification: use one delta for every pair.

Fix $\varepsilon > 0$. Choose $\delta_0 > 0$ for uniform continuity on $[0, K+1]$, and $\delta_1 > 0$ for uniform continuity on $[K, +\infty)$, both for this same tolerance ε. Set

$$\delta = \min\{\delta_0, \delta_1, 1\} > 0.$$

Take $x, y \geq 0$ with $|x - y| < \delta$. Interchanging them if necessary, assume $x \leq y$. If $y \leq K + 1$, both points lie in the first interval and their distance is less than δ_0, so their function values differ by less than ε. If $y > K + 1$, then $y - x < 1$, which rearranges to $x > y - 1 > K$. Thus both points lie in the tail, and their distance is less than δ_1, so the same conclusion holds. These two cases exhaust every pair. Thus f is uniformly continuous on the whole half-line.

(b) Apply part (a) to square root.

The function $\sqrt{x}$ is continuous on $[0, +\infty)$ by Exercise 17.4. Exercise 19.6(b) gives uniform continuity on $[1, +\infty)$. Part (a), with $k = 1$, now proves uniform continuity on $[0, +\infty)$.

Conclusion. A continuous function on the half-line is uniformly continuous whenever it is uniformly continuous on a tail. The square root is one such function.

19.8 Exercise 19.8: The sine difference estimate

What we must show. Use the Mean Value theorem to prove $|\sin x - \sin y| \leq |x - y|$, then deduce uniform continuity of sine on $\mathbb{R}$.

(a) Apply the theorem on the interval between the points.

If $x = y$, both sides are zero. Suppose $x < y$. Sine is continuous on $[x, y]$ and differentiable on (x, y), so the Mean Value theorem gives some $c \in (x, y)$ with

$$\sin y - \sin x = (\cos c)(y - x).$$

Taking absolute values and using $|\cos c| \leq 1$ gives

$$|\sin y - \sin x| = |\cos c|\,|y - x| \leq |y - x|\,.$$

If $x > y$, interchange the points; the absolute values make the desired statement unchanged. This proves the estimate for all pairs of real numbers.

(b) Read off a uniform choice.

Given $\varepsilon > 0$, let $\delta = \varepsilon$. For all $x, y \in \mathbb{R}$ with $|x - y| < \delta$, part (a) gives

$$|\sin x - \sin y| \leq |x - y| < \delta = \varepsilon.$$

The choice is independent of both points.

Conclusion. Sine is uniformly continuous on $\mathbb{R}$, with the global difference bound claimed in (a). In particular, $|\sin t| \leq |t|$ follows by setting $y = 0$.

19.9 Exercise 19.9: A damped oscillation on the whole real line

What we must show. For $f(0) = 0$ and $f(x) = x \sin(1/x)$ when $x \neq 0$, verify continuity, uniform continuity on every bounded subset, and decide uniform continuity on $\mathbb{R}$.

(a) Check the origin as well as the ordinary formula.

At every nonzero point, the reciprocal, sine composition, and product are continuous. At zero,

$$|f(x) - f(0)| \le |x|$$

for $x \neq 0$, since $|\sin(1/x)| \le 1$; the inequality also holds at zero. Thus $\delta = \varepsilon$ proves continuity there. The function is continuous on all of $\mathbb{R}$.

(b) Enclose any bounded set in a closed interval.

Let $S \subseteq \mathbb{R}$ be bounded. Choose $R > 0$ so that $S \subseteq [-R, R]$. The function is continuous on this closed bounded interval, hence uniformly continuous there by Theorem 19.2. Restriction gives uniform continuity on S, even if S is not closed or is empty.

(c), tails: bound the derivative away from the origin.

For $x \neq 0$, the product and chain rules give

$$f'(x) = \sin(1/x) - \frac{\cos(1/x)}{x}.$$

If $|x| \ge 1$, then

$$|f'(x)| \le 1 + \frac{1}{|x|} \le 2.$$

Theorem 19.6 therefore gives uniform continuity on each of the intervals $[1, +\infty)$ and $(-\infty, -1]$. More explicitly, the Mean Value theorem gives the bound $|f(x) - f(y)| \le 2\,|x - y|$ whenever both points lie in either one of these intervals.

(c), combine the tails and the central interval.

The central restriction to $[-2, 2]$ is uniformly continuous by part (b). For a fixed $\varepsilon > 0$, choose its uniform parameter $\delta_0 > 0$ and put

$$\delta = \min\{\delta_0, \varepsilon/2, 1\} > 0.$$

Take $x \le y$ with $y - x < \delta$. If both points lie in $[-2, 2]$, the central estimate gives $|f(x) - f(y)| < \varepsilon$. If $y > 2$, then $x > y - 1 > 1$, so both lie in the positive tail; the derivative bound gives $|f(x) - f(y)| \le 2\,|x - y| < 2\delta \le \varepsilon$. If $x < -2$, then $y < x + 1 < -1$, so both lie in the negative tail, and the same bound applies. These cases cover every pair: a pair outside the central square has either $y > 2$ or $x < -2$. Thus the function is uniformly continuous on $\mathbb{R}$.

This last argument also proves a useful joining fact: a continuous real function that is uniformly continuous on both tails $[1, +\infty)$ and $(-\infty, -1]$ is uniformly continuous on $\mathbb{R}$. In the general case, replace $\varepsilon/2$ above by the minimum of the two tail parameters. The same interval classification is unchanged.

Conclusion. The function is continuous on $\mathbb{R}$, uniformly continuous on every bounded subset, and also uniformly continuous on all of $\mathbb{R}$.

19.10 Exercise 19.10: The quadratic damping has uniform control as well

What we must show. Repeat all three parts of Exercise 19.9 for $g(0) = 0$ and $g(x) = x^2 \sin(1/x)$ when $x \neq 0$.

(a) Verify continuity everywhere.

Away from zero, compositions and products give continuity. At zero,

$$|g(x) - g(0)| \leq x^2.$$

For $\varepsilon > 0$, choose $\delta = \sqrt{\varepsilon}$. Then $|x| < \delta$ makes the displayed error less than ε, so g is continuous at zero as well.

(b) Restrict a continuous function on a larger compact interval.

For any bounded $S \subseteq \mathbb{R}$, choose $R > 0$ containing it in $[-R, R]$. Continuity on that interval implies uniform continuity by Theorem 19.2. Restriction proves uniform continuity on S.

(c) Obtain a derivative bound on all of $\mathbb{R}$.

For $x \neq 0$, differentiation gives

$$g'(x) = 2x \sin(1/x) - \cos(1/x).$$

By Exercise 19.8(a), $|\sin(1/x)| \leq 1/|x|$. Thus $|x \sin(1/x)| \leq 1$ for every $x \neq 0$, and

$$|g'(x)| \leq 2\,|x \sin(1/x)| + |\cos(1/x)| \leq 2 + 1 = 3.$$

The derivative at zero exists too. Indeed, for $h \neq 0$,

$$\left| \frac{g(h) - g(0)}{h} - 0 \right| = |h \sin(1/h)| \leq |h| \longrightarrow 0.$$

This is precisely the difference quotient defining $g'(0)$, so $g'(0) = 0$. Consequently g is differentiable on all of $\mathbb{R}$ and satisfies $|g'| \leq 3$ everywhere.

The Mean Value theorem now gives $|g(x) - g(y)| \leq 3\,|x - y|$ for every pair of real points. The choice $\delta = \varepsilon/3$ proves uniform continuity on $\mathbb{R}$, also as an application of Theorem 19.6.

Conclusion. All three conclusions are affirmative: g is continuous on $\mathbb{R}$, uniformly continuous on every bounded subset, and uniformly continuous on $\mathbb{R}$. Its global derivative bound is 3.

19.11 Exercise 19.11: Uniform continuity of the extended sinc function

What we must show. Accept the continuity on $\mathbb{R}$ of the extension

$$\widetilde{h}(x) = \begin{cases} \sin x/x, & x \neq 0, \\ 1, & x = 0, \end{cases}$$

from Example 9, and prove uniform continuity on $\mathbb{R}$.

1. Bound the derivative on both tails.

For $x \neq 0$, the quotient rule gives

$$\widetilde{h}'(x) = \frac{x \cos x - \sin x}{x^2}.$$

If $|x| \geq 1$, the real triangle inequality yields

$$\left|\widetilde{h}'(x)\right| \leq \frac{|x|\,|\cos x| + |\sin x|}{x^2} \leq \frac{1}{|x|} + \frac{1}{x^2} \leq 2.$$

The function is continuous on each tail and differentiable in its interior, so Theorem 19.6 gives uniform continuity on $[1, +\infty)$ and on $(-\infty, -1]$.

2. Join the two tails through a compact middle interval.

The continuity assumed in the exercise gives uniform continuity on $[-2, 2]$ by Theorem 19.2. Fix $\varepsilon > 0$, let $\delta_0 > 0$ be its central parameter, and set $\delta = \min\{\delta_0, \varepsilon/2, 1\}$. For a pair $x \le y$ with $y - x < \delta$, if both are in $[-2, 2]$, the central estimate works. If $y > 2$, then $x > y - 1 > 1$ and both are in the positive tail. If $x < -2$, then $y < x + 1 < -1$ and both are in the negative tail. On either tail the Mean Value theorem and the bound 2 give an error at most $2\,|x - y| < \varepsilon$. Every pair belongs to one of these cases, which proves the uniform-continuity definition on all of $\mathbb{R}$.

Conclusion. The continuous extension of $\sin x/x$ at zero is uniformly continuous on the whole real line. The argument uses the stipulated continuity at zero and derivative bounds only away from zero.

19.12 Exercise 19.12: The running maximum of a continuous function

What we must show. For a continuous $f : [a, b] \to \mathbb{R}$, prove that

$$f^*(x) = \sup\{f(t) : a \le t \le x\}, \qquad x \in [a, b],$$

is increasing and continuous. Increasing here means nondecreasing, as elsewhere in the text.

1. Verify that every supremum is finite and attained.

For each $x \in [a, b]$, the interval $[a, x]$ is nonempty, closed, and bounded. The restriction of f is continuous on it. The extreme value theorem (Theorem 18.1) therefore supplies some $z \in [a, x]$ with $f(z) = \max_{a \le t \le x} f(t)$. In particular the supremum defining $f^*(x)$ is a real number and is this maximum. When $x = a$, the interval is the singleton $\{a\}$ and the assertion reduces to $f^*(a) = f(a)$.

2. Check the order directly from set inclusion.

If $a \le x \le y \le b$, then $[a, x] \subseteq [a, y]$. Every function value used in the supremum for x is therefore also used for y. Hence $f^*(x) \le f^*(y)$. This proves the required increasing property; it need not be strict, as a constant f demonstrates.

3. Obtain one parameter from uniform continuity of f.

Theorem 19.2 makes f uniformly continuous on $[a, b]$. For $\varepsilon > 0$, choose $\delta > 0$ such that

$$u, v \in [a, b], \quad |u - v| < \delta \quad \Longrightarrow \quad |f(u) - f(v)| < \varepsilon.$$

We prove that this same δ works for f^*.

4. Locate a point attaining the larger running maximum.

Take $x, y \in [a, b]$ with $|x - y| < \delta$. If $x = y$, the two running maxima agree. Otherwise interchange the points if needed so that $x < y$. By Step 1 choose $z \in [a, y]$ with $f(z) = f^*(y)$.

If $z \le x$, then $f(z) \le f^*(x)$, since z is among the points considered in that smaller supremum. Combining this with Step 2 gives $f^*(y) \le f^*(x) \le f^*(y)$, so the two values agree and their difference is zero.

If $z > x$, then $0 < z - x \leq y - x < \delta$. The chosen parameter therefore gives $|f(z) - f(x)| < \varepsilon$. Since $f(x) \leq f^*(x)$ and $f(z) = f^*(y)$, we get

$$0 \leq f^*(y) - f^*(x) = f(z) - f^*(x) \leq f(z) - f(x) \leq |f(z) - f(x)| < \varepsilon.$$

The initial inequality again uses monotonicity from Step 2. These two cases prove $|f^*(y) - f^*(x)| < \varepsilon$ for every eligible pair.

Thus f^* is actually uniformly continuous on $[a, b]$. In particular, fixing either point as an arbitrary x_0 verifies continuity at x_0, including the endpoints. If $a = b$, there is only one point and the same assertion is immediate.

Conclusion. The running supremum is a finite attained maximum, is nondecreasing, and is continuous on $[a, b]$; indeed, it is uniformly continuous there.